



Gesture Control Robotic Hand

Group - 7

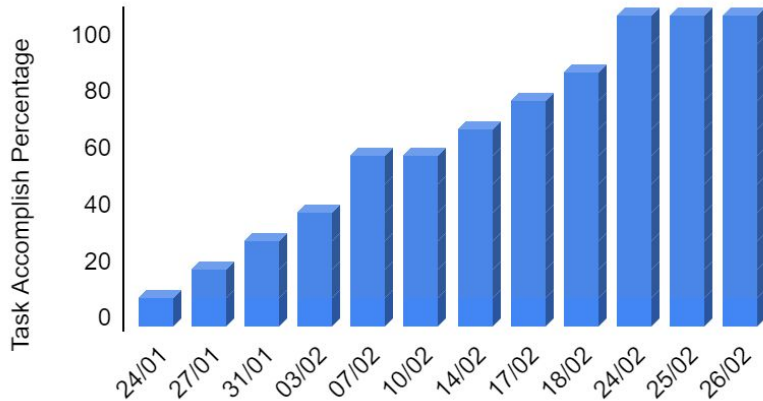
Members

Project Planner

Main Task	24/ 01	27/ 01	31/ 01	03/ 02	07/ 02	10/ 02	14/ 02	17/ 02	18/ 02	24/ 02	25/ 02	26/ 02	Participation				
Brainstorming Ideas	T	T	T	T	T	T	T	T	T	T	T	T	A	B	C		E
Researching Design and Methodology		T	T	T	T	T	T	T	T	T	T	T	A	B	C		E
Preparing Design			T	T	T	T	T	T	T	T	T	T	A	B	C	D	E
Prototype Design 3D print				T	T	T	T	T	T	T	T	T				D	
New Design 3D print					T	T	T	T	T	T	T	T	A			D	
Buying equipment						T	T	T	T	T	T	T	A	B	C	D	E
Training AI					T		T	T	T	T	T	T	A	B	C	D	E
Coding AI Part								T	T	T	T	T	A	B	C	D	E
Coding MCU Part									T	T	T	T	A	B	C	D	E
Assembling Prototype Hands										T	T	T	A	B	C	D	E
3D Printing the final design										T	T	T				D	
Testing final coding and robot											T	T	A	B	C		
Circuit Error Correction											T	T	A	B	C	D	
One Sheet												T	A		C		
Slide												T	A	B	C	D	E

Project Planner

Progress Bar



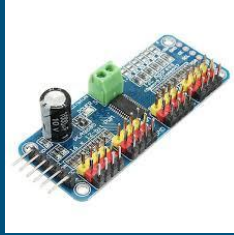
Bill of Materials

Equipment	Qty	Price/Unit	Amount
MG996R Servo Motor	5	260	1300
Servo Driver	1	200	200
Jumper Wire	2	50	100
Servo Tester	1	90	90
Switch	2	110	220
AA6 9 V DC power	1	45	45
Alkaline AA 1.5V(6 Pcs)	1	145	145
Total			2100

Materials



MG996R Servo Motor



Servo Motor Driver



Arduino Uno



Servo Tester



Alkaline AA 1.5 V



Jumper Wire

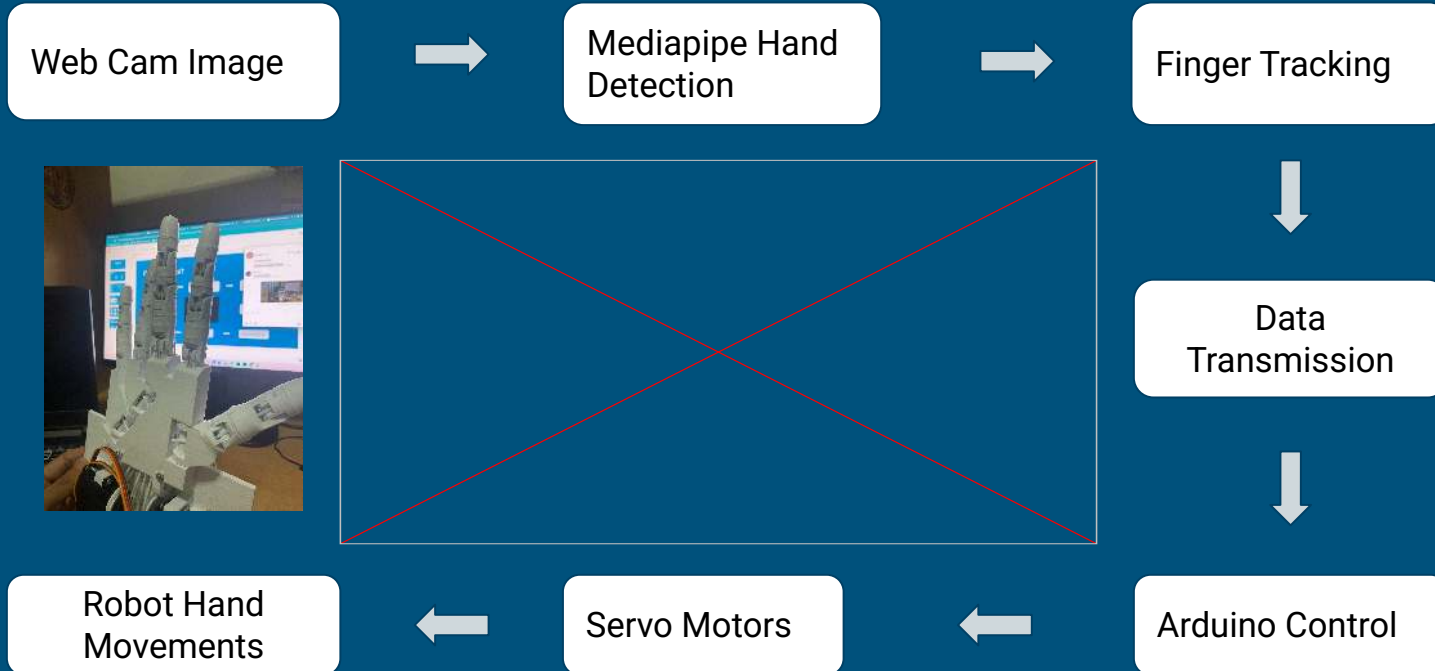


AA6 9V DC Power



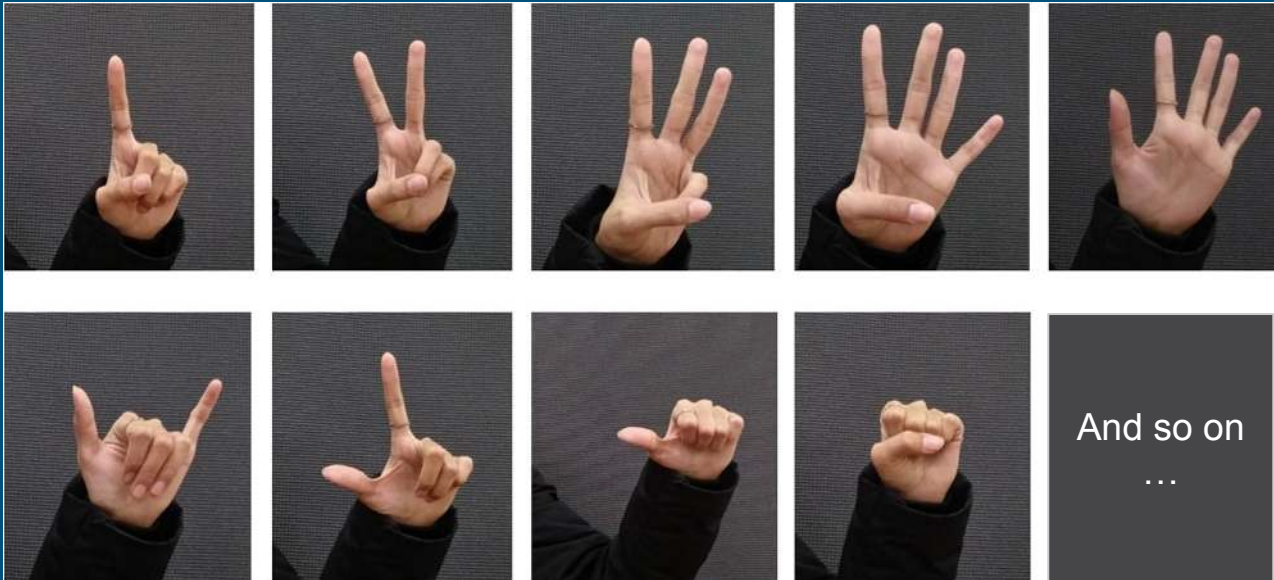
PLA Filament

How it works?



Basic Movements

- The robot hand can imitate 32 (2^5) basic combination of fingers movements.



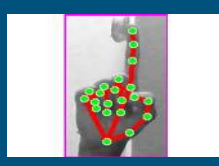
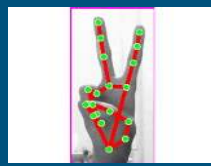
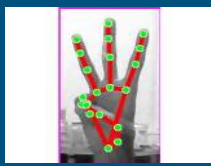
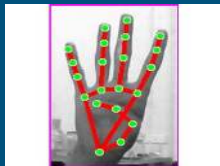
Confidence Score

❖ Image preprocessing

- openCv to Capture Image frame from webcam
- Convert it to grayscale for easier processing (reduce background confusion)
- Use media pipe to detect hand and landmarks to draw on it
- Resize the image to a new square window for consistency
- Saved those images for data training

❖ Teachable Machine

- Train the model using Teachable Machine using the saved data
- Train with epochs 250 setting configuration
- Fed 2000 datasets for each category



Teachable Machine

0

2003 Image Samples

Webcam

Upload



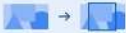
1

2184 Image Samples

File:

Choose images from your files:
or drag & drop here

Import images from
Google Drive


Images will be cropped to square



Training

Train Model

Advanced

Epochs: 250

Batch Size: 16

Learning Rate: 0.001

Reset Defaults

Under the hood

Preview

Export Model

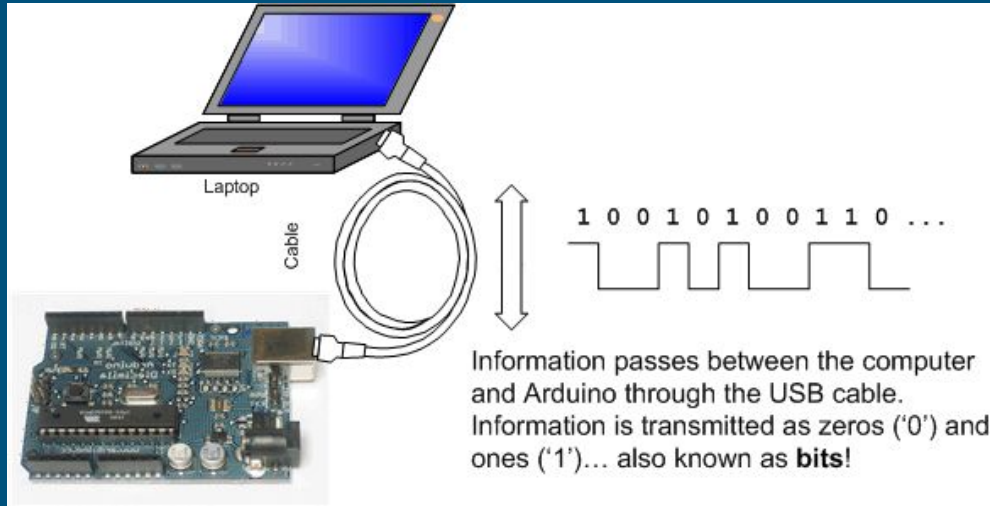
You must train a model on the left before you can preview it here.

Results



Theory Analysis

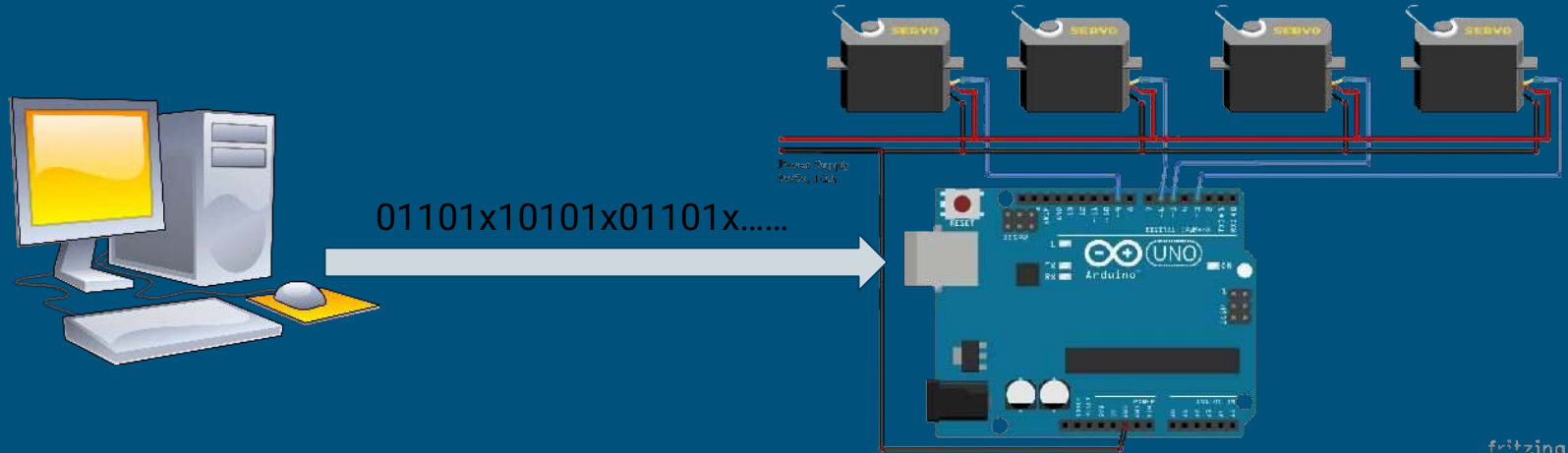
How MCU & Computer talk to each other



1. By using pyserial library, encode and send 5 characters such as [1,0,1,0,1] that were obtained from machine learning algorithm.
2. The baud rate of sending computer and receiving microcontroller should be the same.(9600).

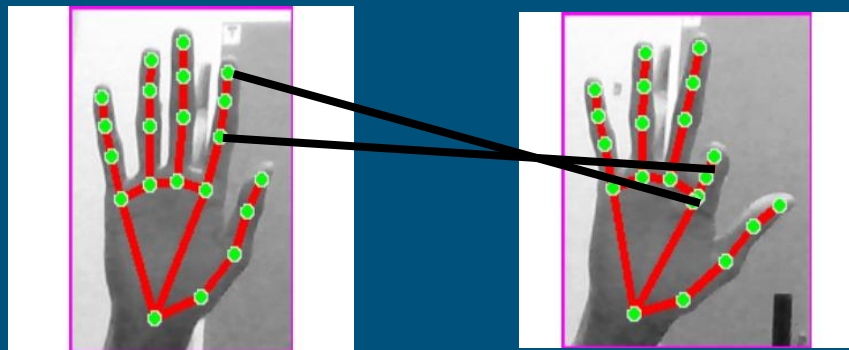
How MCU & Computer talk to each other

We use delimiter character “x” to command MCU to pause receiving data make decisions on which servos to move. Furthermore, we use **OOP for code complexity reduction and speed switching functions**.



Software tools

- ❖ Mediapipe
 - used mediapipe's pretrained models to track the hand and draw the landmarks points on the palm.
 - Using those landmarks, create an algorithm to determine which finger is up or down.



Software tools



- ❖ OpenCv and numpy
 - Used those libraries for image preprocessing purposes.
- ❖ Teachable machine
 - to train our own model for hand gestures and exported as a keras model.
- ❖ Pyserial
 - used it for transmission of data to arduino via serial communication

Error Analysis

Software Errors

- ❖ Why not use teachable machine for robot hand gesture control and why use mediapipe for that?
- ❖ Inaccurate two finger hand gesture recognition?

Hardware Errors

- ❖ Insufficient current for the whole 5 motors to run? (PCA9685)
- ❖ Disconnected serial communication during testing?
- ❖ Friction between joints and not having a smooth control.

Q & A Session

Source code -

